

Suhani Shukla

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Education

University of California — Los Angeles (expected June 2027)

B.S. Computer Science, Minor in Entrepreneurship | GPA: 3.7/4.0

Relevant Coursework: Data Structures & Algorithms, Computer Organization, Operating Systems, Networking, Machine Learning, Deep Learning/Neural Networks, Natural Language Processing, Compiler Design, Cybersecurity, Digital Logic Design

Honors and Awards: National Merit Scholar (2023), UCLA QWER Hacks 3rd Place (Traditional Hack) 2024, AIME Qualifier (2021)

Experience

Machine Learning Software Engineer Intern (Applied AI/ML), Microsoft AI

June 2026 - Present

- Developing Small Language Models (SLMs) for query fanout in Microsoft WebIQ, a Bing Generative Search initiative. Implementing ML experimentation pipelines for teacher model data generation and curation, running controlled ablations across teachers, prompting strategies, query-type distributions, and data recipes for supervised fine-tuning.
- Designed and evaluated reinforcement learning and post-training techniques, including on-policy distillation, direct preference optimization, and reward-based skill optimization. Ran controlled RL experiments across reward formulations and weighting strategies, evaluating effects on checkpoint quality and downstream performance.
- Developing a Gemma-270M regression-based trigger model for dynamic query fanout, predicting the search-quality delta between fanout and original queries; investigating query-text and post-retrieval search metadata and running ablations across labeling thresholds, loss functions, and feature sets to identify predictive signals for fanout utility.
- Improved a critical internal search-quality metric +2.43 points over baseline (3.3%), meeting the target quality threshold for a query-fanout mode, with resulting models selected for production deployment. Building automated evaluation pipelines to measure SLM quality across search tasks, and analyzing model behavior and failure modes across training checkpoints to identify regressions and guide improvements to data and training recipes.
- Benchmarking and implementing SLM serving optimizations, including KV caching and quantization, evaluating latency, memory, and model-quality tradeoffs to reduce inference latency and serving cost for production deployment.

Software Development Engineer Intern, Amazon Web Services

June - Sept. 2025

- Automated end-to-end CI/CD workflows for AWS Redshift Console deployments using Live Pipeline Templates (Ruby) and AWS CDK (TypeScript), migrating legacy infrastructure to reproducible Infrastructure as Code (IaC).
- Onboarded services onto Apollo containers and Application Revisions for deterministic builds and standardized releases across environments. Designed a zero-touch rollback system with real-time canary analysis and composite alarms, minimizing issue blast radius and ensuring safe global deployments across 20+ AWS regions.
- Reduced deployment cycle time from 100+ to <28 days and increased deployment frequency 4x through staged bake times, automated approvals, and promotion blockers. Integrated automated testing, performance monitoring, and deployment safety checks into release pipelines, improving reliability for thousands of enterprise customers globally.

Software Diagnostics Intern, Nokia

June - Aug. 2024

- Developed C++ systems-level diagnostic software for ASIC components, improving reliability of hardware workflows.
 - Implemented bidirectional gRPC streaming for inter-process communication in C++, enabling real-time system-wide logging and diagnostics across all *Diags* test suites. Ensured robustness under prolonged ASIC validation runs.
 - Designed and implemented custom POSIX signal handler to safely manage interrupts (Ctrl-C) for long-running diagnostics. Integrated with Python front-end for improved usability for test engineers and safer orchestration.
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Projects

HR AI Assistant (Production RAG Chatbot)

- Built and deployed a production RAG-based HR assistant for AlphaCore Wealth Advisory using Amazon Bedrock and internal company documentation, launched to 100+ employees.
- Implemented end-to-end application with FastAPI, FAISS, and embeddings-based retrieval, deployed serverless infra on AWS Lambda and S3. Built front-end interface, owning full-stack integration between UI and backend APIs.

EMG-to-QWERTY Keystroke Prediction (Deep Learning Project) 

- Evaluated CNN, LSTM/GRU, Transformer, and CNN-Transformer hybrid models for EMG-based keystroke prediction.
 - Ran controlled experiments across architectures, hyperparameters, and data augmentation strategies including Gaussian noise injection, random masking, and dropout; benchmarked models using Character Error Rate (CER).
 - Analyzed training dynamics, convergence, generalization, and overfitting across architectures to identify model and data configurations that improved sequence prediction.
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Skills

Languages & Systems/Cloud: C++, Python, TypeScript, Docker, AWS, gRPCs, CI/CD, Infrastructure as Code (IaC), AWS CDK, Git
AI/ML: HuggingFace, Transformers, LLMs/SLMs, Post-Training, Supervised Fine-Tuning (SFT), Knowledge Distillation, RL, Training & Evaluation Pipelines, LLM Inference, RAG, Embeddings, CNNs, Azure Machine Learning, Amazon Bedrock